

Algebra II with Trigonometry

Chapter 11.3

Olympiad Problems (GPT-5.4)

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Problems

1. A hyperbola centered at the origin has asymptotes $y = \pm 2x$ and passes through $(3, \sqrt{13})$. Find its standard equation.
2. A hyperbola centered at the origin has foci $(\pm 5, 0)$. One asymptote has slope $\frac{4}{3}$. Find the vertices.
3. A point P lies on a hyperbola with foci $(\pm 13, 0)$, and the absolute difference of its distances to the foci is 10. If P also lies on the line $y = x$, find P .
4. Two hyperbolas are given by

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{and} \quad \frac{y^2}{a^2} - \frac{x^2}{b^2} = 1.$$

If they intersect, determine the coordinates of all intersection points in terms of a and b .

5. A hyperbola centered at the origin has vertices $(\pm 4, 0)$, and one of its asymptotes is perpendicular to the line $3x - 4y = 0$. Find the equation of the hyperbola.
6. Among all hyperbolas centered at the origin with asymptotes $y = \pm x$, one passes through $(5, 3)$ and another through $(13, 12)$. Without graphing,

decide which hyperbola is closer to its center at the vertices, and by how much.